

Physics1 BAS 021

Dr. Mohamed Sobhy

Mohamed.sobhi@buc.edu.eg

Office : 430

Text book: Serway, Jewett - Physics for
Scientists and Engineers with Modern
Physics 9th edition

Course Content:

► Properties of Materials

- * Measurements - units - Dimensional equations - Dimensional Analysis
- * Elasticity - Stress- Strain- Elastic properties of solids
- * Energy - Kinetic energy - Potential energy - Work - Hook's law
- * Oscillatory motion - Simple harmonic motion - Simple Pendulum

► Electricity

Physics and Measurements

Physics

► Fundamental Science

- Concerned with the fundamental principles of the Universe
- Foundation of other physical sciences
- Has simplicity of fundamental concepts

Physics, cont.

► Divided into six major areas:

- Classical Mechanics
- Relativity
- Thermodynamics
- Electromagnetism
- Optics
- Quantum Mechanics

Classical Physics

- ▶ Mechanics and electromagnetism are basic to all other branches of classical and modern physics.
- ▶ Classical physics
 - ▶ Developed before 1900
 - ▶ First part of text deals with Classical Mechanics
 - ▶ Also called Newtonian Mechanics or Mechanics
- ▶ Modern physics
 - ▶ From about 1900 to the present

Objectives of Physics

- ▶ To find the limited number of fundamental laws that govern natural phenomena
- ▶ To use these laws to develop theories that can predict the results of future experiments
- ▶ Express the laws in the language of mathematics
 - ▶ Mathematics provides the bridge between theory and experiment.

Classical Physics Overview

- ▶ Classical physics includes principles in many branches developed before 1900.
- ▶ Mechanics
 - ▶ Major developments by Newton, and continuing through the 18th century
- ▶ Thermodynamics, optics and electromagnetism
 - ▶ Developed in the latter part of the 19th century
 - ▶ Apparatus for controlled experiments became available

Modern Physics

- ▶ Began near the end of the 19th century
- ▶ Phenomena that could not be explained by classical physics
- ▶ Includes theories of relativity and quantum mechanics

Special Relativity

- ▶ Correctly describes motion of objects moving near the speed of light
- ▶ Modifies the traditional concepts of space, time, and energy
- ▶ Shows the speed of light is the upper limit for the speed of an object
- ▶ Shows mass and energy are related

Quantum Mechanics

- ▶ Formulated to describe physical phenomena at the atomic level
- ▶ Led to the development of many practical devices

Measurements

- ▶ Used to describe natural phenomena
- ▶ Each measurement is associated with a physical quantity
- ▶ Need defined standards
- ▶ Characteristics of standards for measurements
 - ▶ Readily accessible
 - ▶ Possess some property that can be measured reliably
 - ▶ Must yield the same results when used by anyone anywhere
 - ▶ Cannot change with time

Standards of Fundamental Quantities

- ▶ Standardized systems

- ▶ Agreed upon by some authority, usually a governmental body

- ▶ SI - Système International

- ▶ Agreed to in 1960 by an international committee
 - ▶ Main system used in this text

Fundamental Quantities and Their Units

Quantity	SI Unit
Length	meter
Mass	kilogram
Time	second
Temperature	Kelvin
Electric Current	Ampere
Luminous Intensity	Candela
Amount of Substance	mole

Quantities Used in Mechanics

- ▶ In mechanics, three fundamental quantities are used:
 - ▶ Length
 - ▶ Mass
 - ▶ Time
- ▶ All other quantities in mechanics can be expressed in terms of the three fundamental quantities.

Length

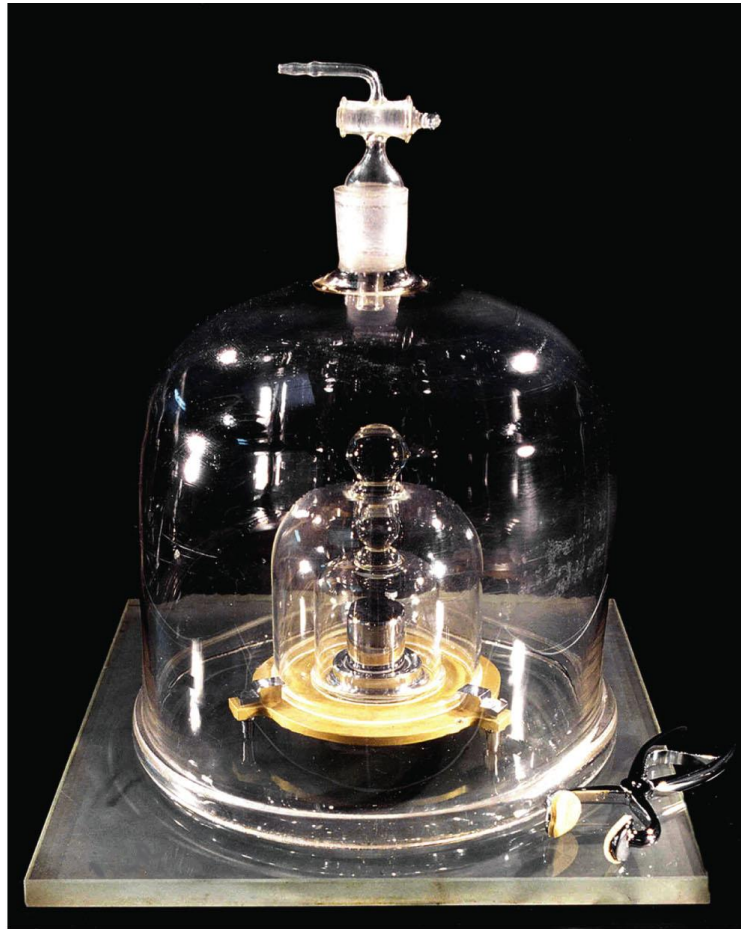
- ▶ Length is the distance between two points in space.
- ▶ Units
 - ▶ SI - meter, m
- ▶ Defined in terms of a meter - the distance traveled by light in a vacuum during a given time

Mass

► Units

- SI - kilogram, kg
- Defined in terms of a kilogram, based on a specific cylinder kept at the International Bureau of Standards
- See Table 1.2 for masses of various objects.

Standard Kilogram



a

Section 1.1

Time

- ▶ Units

- ▶ seconds, s

- ▶ Defined in terms of the oscillation of radiation from a cesium atom

- ▶ See Table 1.3 for some approximate time intervals.

US Customary System

- Still used in the US, but text will use SI

Quantity	Unit
Length	foot
Mass	slug
Time	second

Prefixes

- ▶ Prefixes correspond to powers of 10.
- ▶ Each prefix has a specific name.
- ▶ Each prefix has a specific abbreviation.
- ▶ The prefixes can be used with any basic units.
- ▶ They are multipliers of the basic unit.
- ▶ Examples:
 - ▶ $1 \text{ mm} = 10^{-3} \text{ m}$
 - ▶ $1 \text{ mg} = 10^{-3} \text{ g}$

Prefixes, cont.

TABLE 1.4 *Prefixes for Powers of Ten*

Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{-24}	yocto	y	10^3	kilo	k
10^{-21}	zepto	z	10^6	mega	M
10^{-18}	atto	a	10^9	giga	G
10^{-15}	femto	f	10^{12}	tera	T
10^{-12}	pico	p	10^{15}	peta	P
10^{-9}	nano	n	10^{18}	exa	E
10^{-6}	micro	μ	10^{21}	zetta	Z
10^{-3}	milli	m	10^{24}	yotta	Y
10^{-2}	centi	c			
10^{-1}	deci	d			

Fundamental and Derived Units

▶ Derived quantities can be expressed as a mathematical combination of fundamental quantities.

▶ Examples:

- ▶ Area

- ▶ A product of two lengths

- ▶ Speed

- ▶ A ratio of a length to a time interval

- ▶ Density

- ▶ A ratio of mass to volume

Basic Quantities and Their Dimension

- ▶ Dimension has a specific meaning - it denotes the physical nature of a quantity.
- ▶ Dimensions are often denoted with square brackets.
 - ▶ Length [L]
 - ▶ Mass [M]
 - ▶ Time [T]

Dimensions and Units

- ▶ Each dimension can have many actual units.
- ▶ Table 1.5 for the dimensions and units of some derived quantities

TABLE 1.5 *Dimensions and Units of Four Derived Quantities*

Quantity	Area (A)	Volume (V)	Speed (v)	Acceleration (a)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

Dimensional Analysis

- ▶ Technique to check the correctness of an equation or to assist in deriving an equation
- ▶ Dimensions (length, mass, time, combinations) can be treated as algebraic quantities.
 - ▶ Add, subtract, multiply, divide
- ▶ Both sides of equation must have the same dimensions.
- ▶ Any relationship can be correct only if the dimensions on both sides of the equation are the same.
- ▶ Cannot give numerical factors: this is its limitation

Dimensional Analysis, example

- ▶ Given the equation: $x = \frac{1}{2} at^2$
- ▶ Check dimensions on each side:

$$L = \frac{L}{\cancel{T^2}} \cdot \cancel{T^2} = L$$

- ▶ The T^2 's cancel, leaving L for the dimensions of each side.
 - ▶ The equation is dimensionally correct.
 - ▶ There are no dimensions for the constant.

- Show that the following equation is dimensionally correct:

$V = V_0 + at$ where (V_0) is velocity and (a) is acceleration

- Solution: $Dim[LHS] = [L/T]$

$$Dim[RHS] = [L/T] + [L / T^2][T] = [L/T] + [L/T]$$

$Dim[LHS] = Dim[RHS] \Rightarrow \therefore$ The equation is dimensionally correct (ignore constants)

Newton's second law states that acceleration is proportional to the force acting on an object and is inversely proportional to the object mass. What are the dimensions of force?

Solution: *The acceleration $a \rightarrow [a] = [L]/[T]^2$, The force $F \rightarrow [F]??$*

The mass $m \rightarrow [m] = [M]$

$$a \propto Fm^{-1} \Rightarrow a = kF/m$$

$$\therefore F = ma/k \Rightarrow [F] = [m][a] = [M][L][T^{-2}]$$

Dimensional Analysis to Determine a Power Law

► Determine powers in a proportionality

► Example: find the exponents in the expression

$$x \propto a^m t^n$$

► You must have lengths on both sides.

► Acceleration has dimensions of L/T^2

► Time has dimensions of T .

► Analysis gives

$$x \propto at^2$$